Date: ________ Day: ________

Testing Overall Regression Model

Hypothesis:

$H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 \cdots \beta_k = 0$

(None of variable explain Y)

H_1 : At least one of the β_j is not zero

Test Statistics:

$$F = \frac{SSR/k}{SSE/N-k-1} = \frac{MSR}{MSE}$$

K : Number of variables

N : Number of obs.

$SS\cancel{R}E = \Sigma (y_i - \hat{y}_i)^2$

$SS\cancel{E}R = \Sigma (\hat{y} - \bar{y})^2$

Critical Region:

$F > F_\alpha (k, N-k-1)$

ERROR:
V1= 1
BUT K ! = 1
K= NUMBER OF VARIABLES IN FULL MODEL

Date:________ Day:________

Individual variable Screening using F-Test

if we are doing for β_1 then

Hypothesis:

$H_0 : \beta_1 = 0$

$H_1 : \beta_1 \neq 0$

Test Stats:

$$F = \frac{R(\beta_1 \mid \beta_2 \dots \beta_K)}{S^2}$$

$$S^2 = \frac{SSE}{N-K-1} \quad [K=1$$

$$R(\beta_1 \mid \beta_2 \dots \beta_K) = \underbrace{SSR_{full}}_{\text{with } x_1} - \underbrace{SSR_{red}}_{\text{without } x_1}$$

Critical Region:

$$F > F_\alpha(1, N-K-1)$$

$\boxed{K=1}$

ERROR:
V1= 2
BUT K ! = 2
K= NUMBER OF VARIABLES IN FULL MODEL

Date: Day:

Pair Testing:

Hypothesis:

$$H_0 : \beta_1 = \beta_2 = 0$$

$$H_1 : \beta_1 \text{ and } \beta_2 \neq 0$$

Test Statistics

$$F = \frac{R(\beta_1, \beta_2 \mid \beta_3, \beta_4 \ldots \beta_K)/2}{S^2}$$

$$S^2 = \frac{SSE}{N-K-1}$$

$$\boxed{K = 2}$$

Critical Region:

$$F > F_\alpha(2, N-K-1)$$

$$K = 2$$

Date: ________ Day: ________

Coefficient of Determination:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

- Useful when $K = 1$
- $R^2 \times 100\%$ = Percentage of variation.
- Never decrease by adding variables.

$$R^2_{\text{Adjusted}} = 1 - \frac{SSE/(N-K-1)}{SST/(N-1)}$$

- Useful when $K > 1$
- Adjusts R^2 for No. of variables
- Can decrease if adding variables

EXAMPLE: Test whether the overall regression model is statistically significant at the 5% level.

Table 12.3: Data for Example 12.4

y (% survival)	x_1 (weight %)	x_2 (weight %)	x_3 (weight %)
25.5	1.74	5.30	10.80
31.2	6.32	5.42	9.40
25.9	6.22	8.41	7.20
38.4	10.52	4.63	8.50
18.4	1.19	11.60	9.40
26.7	1.22	5.85	9.90
26.4	4.10	6.62	8.00
25.9	6.32	8.72	9.10
32.0	4.08	4.42	8.70
25.2	4.15	7.60	9.20
39.7	10.15	4.83	9.40
35.7	1.72	3.12	7.60
26.5	1.70	5.30	8.20

Hypothesis:

$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$

H_1 : at least one of the β_j is not zero

Test stats:

$$F = \frac{SSR/K}{SSE/N-K-1}$$

$$\begin{bmatrix} N = 13 \\ K = 3 \end{bmatrix}$$

$F = 30.98$

Critical Region:

$\alpha = 0.05$ $\quad F_\alpha(K, N-K-1)$

$K = 3$

$N-K-1 = 9$

$F_{0.05}(3, 9) = 3.86$

Reject H_0

At least one of the β_js is not zero.

EXAMPLE: Using the F-test, determine whether the variables X2 and X3 significantly contribute to the model. Use α=0.05

Table 12.3: Data for Example 12.4

y (% survival)	x_1 (weight %)	x_2 (weight %)	x_3 (weight %)
25.5	1.74	5.30	10.80
31.2	6.32	5.42	9.40
25.9	6.22	8.41	7.20
38.4	10.52	4.63	8.50
18.4	1.19	11.60	9.40
26.7	1.22	5.85	9.90
26.4	4.10	6.62	8.00
25.9	6.32	8.72	9.10
32.0	4.08	4.42	8.70
25.2	4.15	7.60	9.20
39.7	10.15	4.83	9.40
35.7	1.72	3.12	7.60
26.5	1.70	5.30	8.20

ERROR:

V2 = N-K-1 = 13-3-1

Hypothesis:

$H_0 : \beta_2 = \beta_3 = 0$

$H_1 : \beta_2$ and β_3 are not zero

Test stats:

$$F = \frac{R(\beta_2, \beta_3 \mid \beta_1)/2}{S^2}$$

$$= 24.68$$

Critical Region:

$\alpha = 0.05$

$N = 13$

$K = 2$

$N-K-1 = 13-2-1 = 10$

$F_{0.05}(2, 10) = 4.10$

$24.68 > 4.10$

Reject H_0

β_2, β_3 are not zero.